

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

OCTOBER EXAMINATION

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

MONDAY, 21 OCTOBER 2024

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

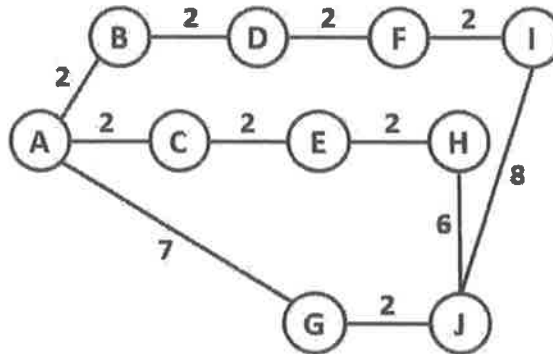
Question 1**Figure 1-1.** Road map of state X

Figure 1-1 shows a road map of 10 different cities represented by nodes A to J in state X, where the label of every arc that connects two nodes represents the distance between them in kilometres, and the straight-line distance between node J and all nodes including node J itself is given in **Table 1-1**.

Table 1-1: Straight-line distance (in kilometres) between node J and all nodes

A	B	C	D	E	F	G	H	I	J
6	6	5	5	3	4	2	3	4	0

Suppose *Breadth-First Search (BFS)*, *Depth-First Search (DFS)* and *A* Search (A*)* are three search strategies implemented in search of the *shortest path from node A to node J*, where their configurations are provided in **Table 1-2**, answer the following **THREE (3)** questions:

Table 1-2: Configuration of the three search strategies

Configuration	BFS	DFS	A*
Graph search implementation	✓	✓	✓
Early goal test implementation	✓		
Late goal test implementation		✓	✓
Node Expansion in alphabetical order	✓	✓	✓

- Classify *BFS*, *DFS* and *A** as either *uninformed search* or *informed search*. (2 marks)
- Provide **TWO (2)** similarities between *uninformed search* and *informed search*. (2 marks)
- For each of the **THREE (3)** search strategies,
 - Construct a *resulting search tree*. (12 marks)
 - Identify the *sequence of node expansion*. (3 marks)
 - Identify the *solution returned*. (3 marks)
 - Compute the *cost of the returned solution*. (3 marks)

[Total: 25 marks]

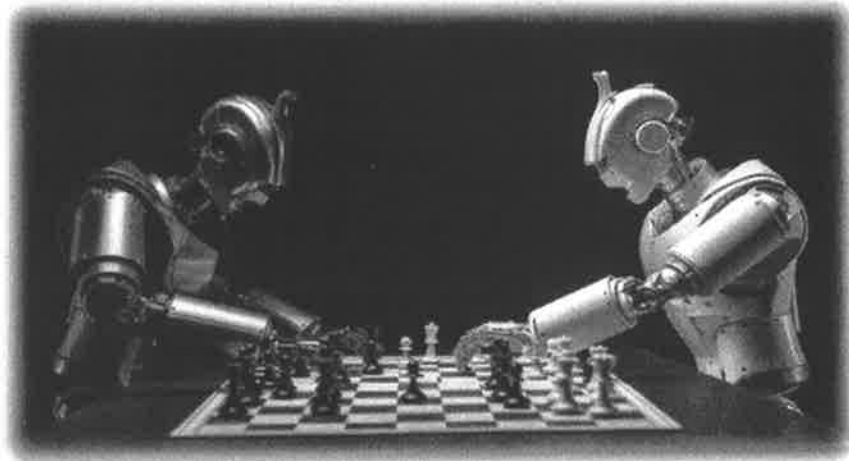
Question 2

Figure 2-1. Two chess agents are playing chess against each other

- a) *Game playing* is a popular application of Artificial Intelligence (A.I.) that involves development of computer programs that can play games such as *chess* (see **Figure 2-1**). These programs process *states* and *actions* from time to time so that they can reason about what to do next and make decisions that will lead to winning outcomes.
- (i) *Agent* is a term that is commonly used to refer to computer programs that are equipped with some kind of intelligence. Explain the meaning of this term in your own words and illustrate your explanation using the diagram provided in **Figure 2-2**. (2 + 3 marks)

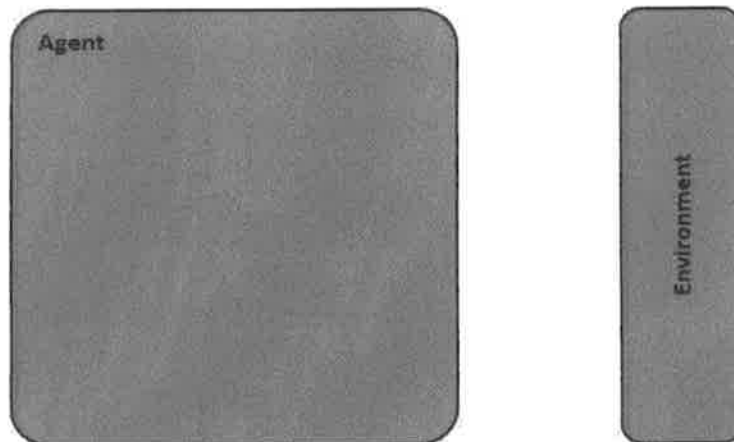


Figure 2-2. Agent and environment

- (ii) For most of the games in A.I., a different term is often used to refer to an *action* as well as a *state*. Identify these two terms. (2 marks)
- (iii) Games that are most studied in A.I. usually have **FIVE (5)** common characteristics: *two-player*, *turn-taking*, *perfect information*, *deterministic*, and *zero-sum*. Briefly explain the meaning of each of these characteristics. (5 marks)
- b) *Game tree search* in A.I. involves exploring all possible moves and outcomes of a game to make optimal decisions. Answer the following questions (refers to next page) based on the game tree given in **Figure 2-3**.

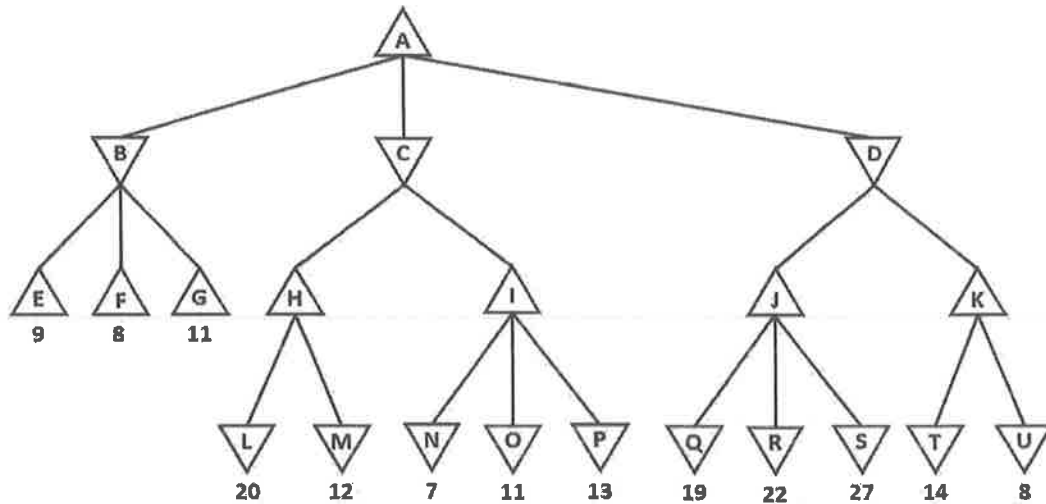
Question 2 b) (Continued)

Figure 2-3. A partial game tree that illustrates the game play of a chess game

- (i) A game illustrated by the game tree provided in **Figure 2-3** indicates that the *environment* is a *multi-agent*. Identify and briefly describe what type of *multi-agent environment* is this. (1 + 1 marks)
- (ii) Suppose minimax algorithm is used to explore the game tree in search of an optimal move, compute the *minimax value* for nodes A, B, C, D, H, I, J, and K. (4 marks)
- (iii) Identify the best move for the MAX player at node A. Justify your answer. (1 + 1 marks)
- (iv) Suppose *minimax search* is applied to the game tree with *alpha-beta pruning* performed in left-to-right order, deduce the final *alpha's* and *beta's* values for nodes A, B, C, D, H, I, J and K. Note that you are required to write the pair of *alpha* and *beta* values for each node in the format of $[\alpha, \beta]$. Then, deduce nodes that would be pruned. (5 marks)

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 3**

FingerTec Face Recognition Face ID 4 is a facial recognition system (see **Figure 3-1**) designed by a local company to automate daily sign-in procedure of a company. To make this works, a high-quality video camera can be installed at the reception area facing the entrance of an office to scan the face of staff so that the system can recognise and identify their identities as they enter.



Figure 3-1. Facial recognition technology

- a) *PEAS*, stands for *Performance, Environment, Actuator* and *Sensor*, describes how an intelligent agent works. Suppose the lighting of the reception area is turned on 24/7 and adjusted to optimise the performance of the *facial recognition system*, and all users are aware of the need of facing directly into the video camera to increase its chances of success, construct a *PEAS* description for the problem to which the system is going to solve. (4 marks)
- b) Identify **SIX (6)** properties of the *facial recognition system*'s environment based on information given below and justify your answer. (6 + 6 marks)
- Fully observable, partially observable or non-observable
 - Episodic or sequential
 - Single-agent or multi-agent
 - Static or dynamic
 - Deterministic or non-deterministic
 - Discrete or continuous (in terms of percept)
- c) Suppose the *facial recognition system* is a *learning agent* consisting of some learning elements, illustrates how a *learning agent* works using the diagram given in **Figure 3-2**. (9 marks)

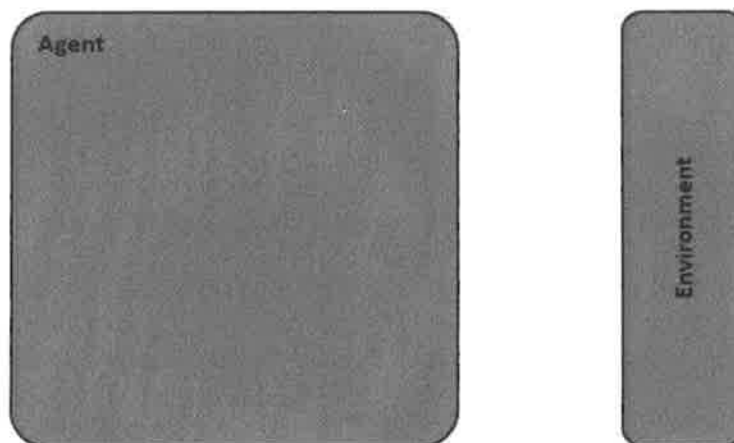


Figure 3-2. A learning agent and its environment

[Total: 25 marks]

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4**

Knowledge-based agents (see **Figure 4-1**) are systems that employ a comprehensive collection of knowledge which can be used to reason, make decisions and solve problems. Usually, they are composed of **TWO (2)** important elements, a *knowledge base* and an *inference system*, that enable them to perform tasks which seems impossible for problem-solving agents. A *knowledge base* can be structured using *propositional logic* which is a logical representation that uses *propositions*, *logical operators*, and *truth values*.

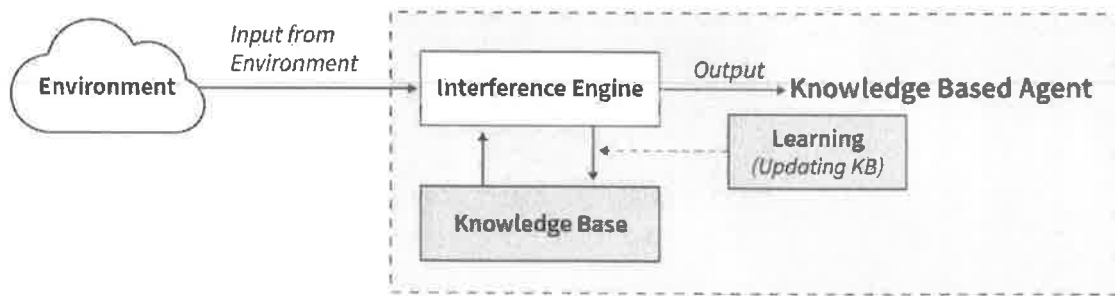


Figure 4-1. A knowledge-based agent (source of image: scaler.com)

- Outline **THREE (3)** capabilities of *knowledge-based agents* that are absent from *problem-solving agents*. (3 marks)
- Briefly describe *propositions* and *logical operators*. (2 marks)
- Translate the following 10 English sentences into *propositions* based on information provided in **Table 4-1**. (10 marks)

Table 4-1: Propositions and their symbols

Symbol	Proposition
A	I will drink coffee.
B	OldTown White Coffee is open.
C	I am happy.
D	I will drink tea.
E	KFC is closed.

- I am not happy, and I will not drink tea.
- KFC is closed or KFC is not closed.
- If OldTown White Coffee is open, then I will drink coffee.
- I will drink tea whenever I am not happy.
- It is not the case that, if I am not happy, then I will not drink coffee or tea.
- I will either drink coffee or tea.
- If OldTown White Coffee is not open and KFC is closed, then I am not happy and I will drink coffee but not tea.

AACS3273 FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**Question 4 c) (Continued)**

- (viii) I will drink coffee if and only if OldTown White Coffee is open and I am happy.
 - (ix) If I am not happy, then I will not drink coffee, but I will drink tea.
 - (x) It is not the case that, if OldTown White Coffee is not open and KFC is not closed, then I am not happy and I will not drink both coffee and tea.
- d) Briefly explain the meaning of *truth table*. Then, construct a *truth table* for each of the following propositions and identify if it is a *tautology*. (1 + 9 marks)
- (i) The proposition created as the answer to **Question 4 c) (i)**.
 - (ii) The proposition created as the answer to **Question 4 c) (ii)**.
 - (iii) The proposition created as the answer to **Question 4 c) (iii)**.

[Total: 25 marks]